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ENBC 455 Bioinformatics Engineering

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Study-A-Gene Assignment

During the summer transition from sophomore to junior year in high school, I participated in the Wearable Tech Ventures Global Youth Hackathon. The challenge was to develop and present a wearable medical device concept that can specifically benefit those diagnosed with Alzheimer's. Although I never understood the technical biology behind how it occurs, I knew this disease put many patients through devastating physical, intellectual, and emotional adversities. Now, I am taking a bioinformatics engineering course that has offered me the opportunity to learn more about this dangerous type of dementia and what can influence its severity.

According to the National Institute on Aging, a subdivision of the National Institute of Health, "One well-known gene that influences Alzheimer's risk is the apolipoprotein E (APOE) gene," ("Alzheimer's Disease Genetics Fact Sheet") and has genetic variants called alleles. The one specific APOE increasing the risk of Alzheimer's is the APOE ϵ 4 with other variations having a neutral or protection effect. To develop a further understanding of its genomic information, I searched for the APOE gene in the National Library of Medicine's National Center for Biotechnology Information (NCBI) gene library. This gene is located between base pairs 44905796 and 44909393 in chromosome 19, built from 6 total introns and 5 total exons. Additionally, the APOE gene consists of 5 alternative spliced transcripts.

Figure 1: Gene Structure and Transcripts of APOE from NBCI

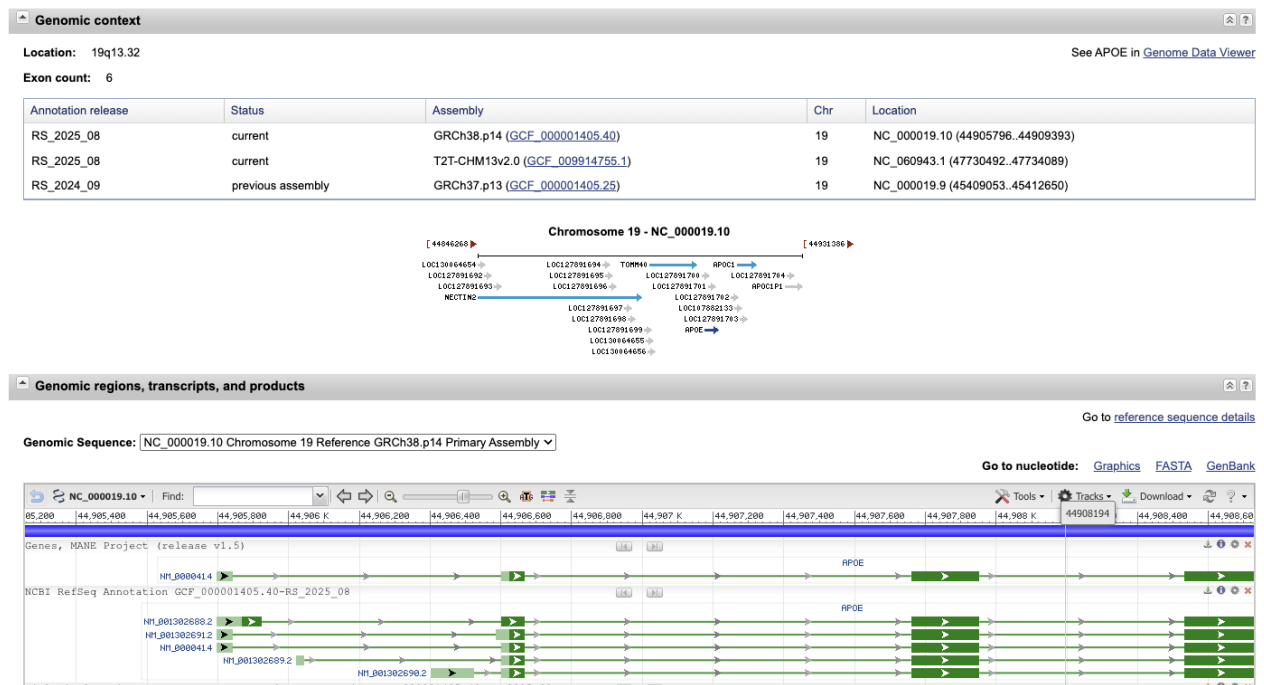


Fig. 1. Noticeably, in the Genomic regions, transcripts, and products section under the NCBI RefSeq Annotation GCF_000001405.40-RS_2025_08, it is shown that the bottom two sequences have different start sites, indicating the APOE gene has multiple promoters. The 3 existing paralogs to the APOE gene can be found in the diagram with light blue colored labeled arrows, e.g. the APOC1 gene. In other species, orthologs of the APOE gene exist mostly in mammals and occasionally fish or amphibians. Examples include but are not limited to the house mouse, norway rat, chimpanzee, domestic cattle, pig, rabbit, dog, tropical clawed frog, torafugu, etc.

The primary function of the APOE gene is “making a protein that helps carry cholesterol and other types of fat in the bloodstream.” (“Alzheimer's Disease Genetics Fact Sheet”)

Transportation of cholesterol is vital due to its role in the creation of cell membranes, hormones, and vitamin D. The same principle applies to fat transportation and its impact on the overall

energy consumption from organs. APOE is described similarly about its functions on the NCBI website with how it “enables cholesterol transfer activity”, “enables enzyme binding”, “enables lipid binding”, “involved_in_cholesterol metabolic process”, etc. (NCBI APOE)

Figure 2: APOE Gene Expression in Different Tissues.

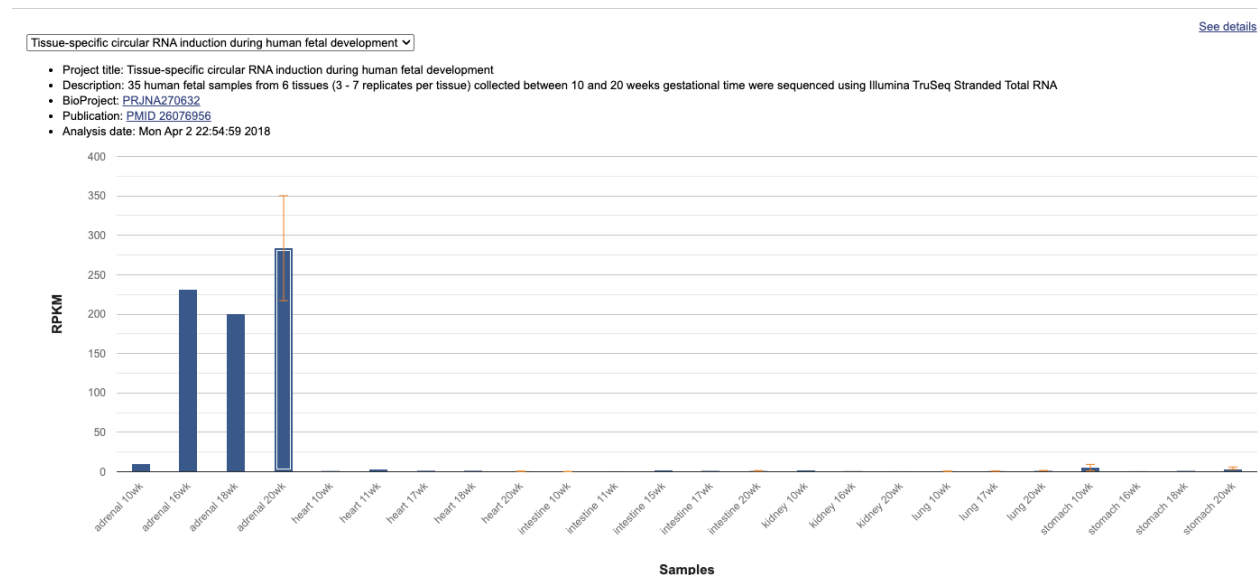
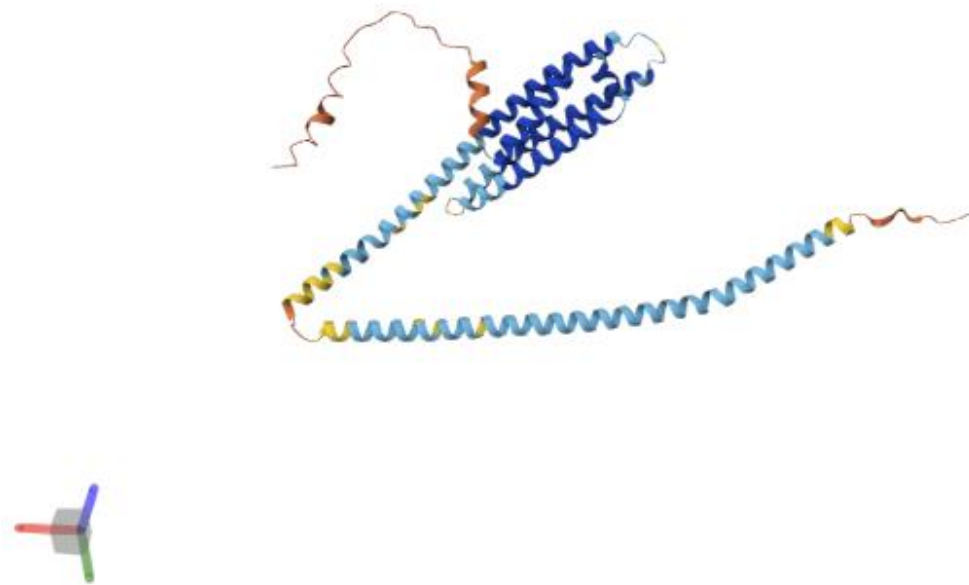


Fig. 2. The x-axis of the graph represents the different types of tissues, organs, or cell types. The y-axis depicts the expression level, quantity of mRNA, that exists in the corresponding tissue, organ, or cell type. Based on visual observation, the adrenal glands seem to consist of the highest expression levels compared to other referenced tissues, organs, and cell types.

As observed from Fig. 2, out of all the tissues, organs, or cell types the APOE gene existed in, Adrenal glands consisted of the highest expression levels. Adrenal glands are “small, triangle-shaped glands that are located on top of each of your two kidneys. They’re a part of your

endocrine system and produce certain hormones that help regulate several important bodily functions” (“Adrenal Gland: What it is, Function, Symptoms & Disorders”).

Figure 3: APOE Visual Protein Structure in Homo sapiens (Human) by Google Deepmind Alphafold Prediction Model



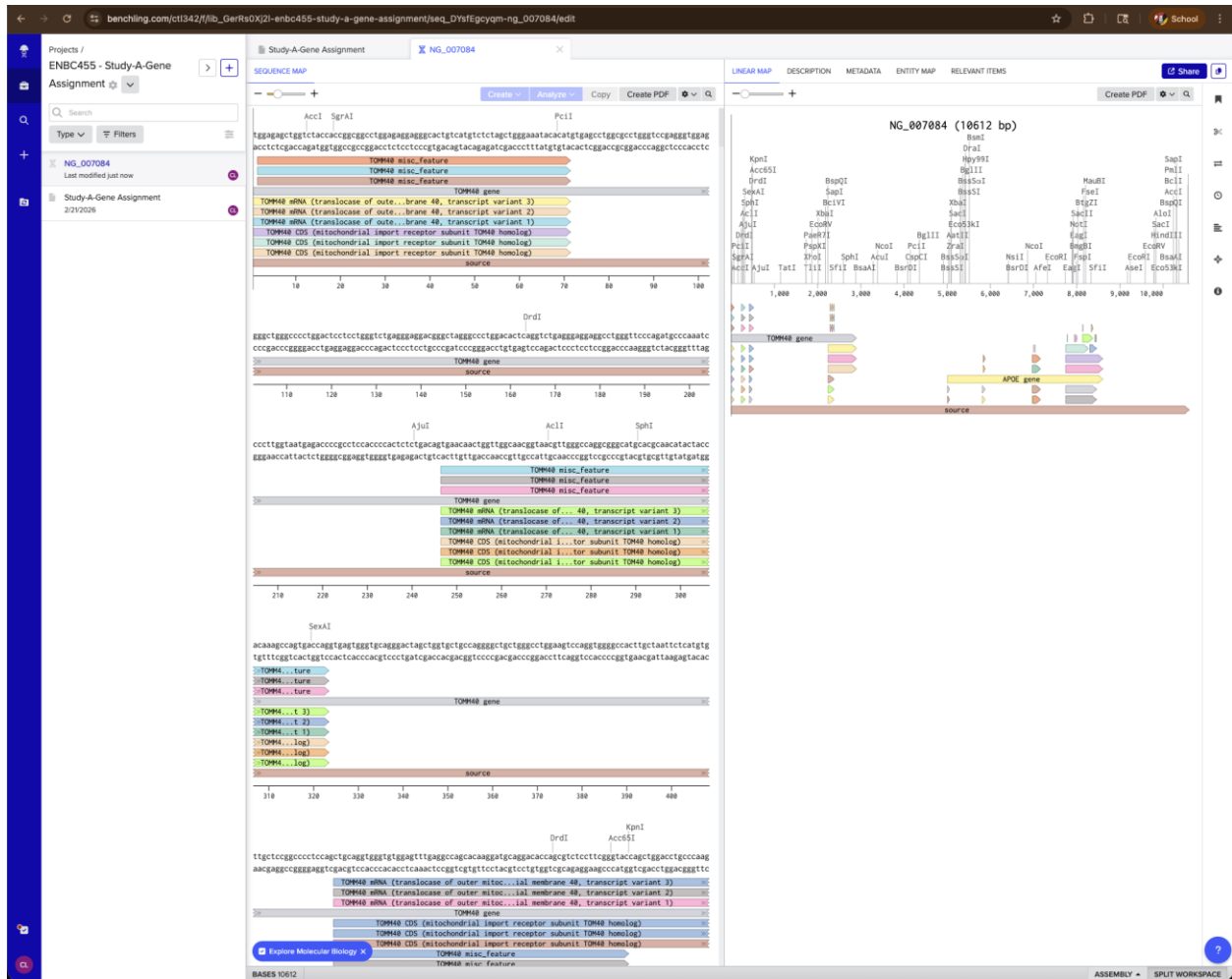
Currently, the APOE gene has a total of 1833 Single Nucleotide Polymorphisms (SNP) which is the total number of variations at a single, specific nucleotide position from the original reference APOE gene sequence. Clinically relevant SNPs describe SNPs specific to a certain species. More specifically, 289 SNPs exist that are clinically relevant for Homo Sapiens (Humans). Among all 289 clinically relevant SNPs, I chose to examine the APOE gene variant

p.Gly183Ala (Variation ID: 1341575), associated with Familial type 3 hyperlipoproteinemia.

Essentially, this condition prohibits the body from being able to break down an accumulation of cholesterol within the bodies' blood stream leading to very serious health conditions. The mutation is a substitution from a G to C nucleotide at position 44,908,844.

The key findings of most of the information related to the apolipoprotein E (APOE) gene were sourced from the NCBI Gene database along with its key tools. More specifically, the NCBI ClinVar was used to identify and filter pathogenic SNPs and the NCBI's Gene Ontology (GO) annotations to categorize the protein's roles. The AlphaFold Protein Structure database was also utilized to obtain a clear 3D model of the APOE protein structure. Benchling, even though it is not a database itself, imported the RefSeqGene sequence to obtain a more detailed visualization of the annotated exons, and introns (refer to appendix A).

APOE Gene RefSeqGene Sequence with Annotations Uploaded to Benchling



Appendix B

APOE Gene RefSeqGene Sequence in FASTA Format

>NG_007084.2 Homo sapiens apolipoprotein E (APOE), RefSeqGene on chromosome 19; nuclear gene for mitochondrial product

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Appendix C

APOE Gene RefSeq Transcript Sequence in FASTA Format

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Appendix D

APOE Gene RefSeq Protein Sequence in FASTA Format

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